

Joe Wallace

Computational Scientist

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Profile

Computational scientist working across molecular and interfacial modelling, scientific Python, and experiment-linked data analysis. I build reproducible workflows for physically grounded modelling, sparse-data analysis, and interpretable scientific decision support across chemistry, materials, and molecular systems.

Core Skills

- Molecular, nanoscale, and interfacial modelling
- Scientific Python, data analysis, and reproducible technical workflows
- Sparse and multivariate scientific data analysis
- Mechanism-aware modelling linked to experimental questions

Experience

Postdoctoral Researcher, Istituto Italiano di Tecnologia, Genova Current
Molecular simulation, data analysis, and experiment-linked scientific computing for nanoscale and interfacial systems.

Doctoral Researcher, University of Liverpool PhD completed 2022
Research on peptide-functionalised gold nanoparticles using experimental characterisation, molecular modelling, molecular dynamics, and enhanced sampling.

Selected Work

- Modelling PET nanoplastics, including contaminant binding, colloidal behaviour, zeta potential, and aggregation.
- Computational prioritisation of functional monolayers on gold nanoparticles for selective molecular recognition.
- Python-based analysis workflows for molecular simulation, small experimental datasets, and interpretable model comparison.
- Exploratory technical work in time-series prediction, ARC-style reasoning tasks, and simple simulation models.

Education

PhD, University of Liverpool 2022
Thesis: *How Does Constraining Folded Peptides within Dense Self-Assembled Monolayers on Gold Nanoparticles Affect Their Structure?*

Selected Publications

Wallace, J., Riccardi, L., Mancin, F., & De Vivo, M. (2025). A scoring function for monolayer-protected gold nanoparticles capable of recognizing small organic molecules in solution. *Journal of Chemical Theory and Computation*.

Franco-Ulloa, S., Cesari, A., Zanoni, G., Riccardi, L., **Wallace, J.**, et al. (2025). Rational design of gold nanoparticle-based chemosensors for detection of the tumor marker 3-methoxytyramine. *Chemical Science*.

Technical Keywords

Molecular modelling Interfacial modelling Scientific Python Data analysis Simulation workflows